

45. Decryption On The Fly (DOTF)

45.1 Overview

DOTF has a function to decode read data of the AXI bus using AES core.

Table 45.1 lists the DOTF specifications and for a system using DOTF, see [section 44.1. Overview](#).

Table 45.1 DOTF Specification

Item	Description
Number of channels	2 channels DOTF0 decodes the read data for OSPI0 and DOTF1 decodes the read data for OSPI1.
Clock Source	Register clock : PCLKB AES core clock : PCLKA
AES core function	- Utilize for on-the-fly decryption of encrypted software stored in external memory - Block size: 128-bit - Key size: 128-bit, 192-bit, 256-bit - Support the following block cipher mode: - Counter (CTR) mode following NIST SP800-38A - Supports side channel countermeasure function - Supports self-test function - Counter[127:0] = {IV[127:28], Address[31:4]}, where Address is the memory mapped address of the encrypted data, aligned to the AES block size of 128 bits (16 bytes). The IV is chosen according to the recommendations provided in Appendix B of SP800-38A. See section 44, Octal Serial Peripheral Interface (OSPI) and section 5, Address Space for details. See NIST SP800-38A for details of AES counter mode operation.
Tamper Resistance	Countermeasures available for side-channel attacks, including SPA/DPA and timing attacks
Module-stop function	Module-stop state can be set to reduce power consumption (same as OSPI module stop)
TrustZone Filter	Security and Privilege attribution can be set for each channel (same as TZF of OSPI)

45.2 Block Diagram

The block diagram of DOTF is shown in [Figure 45.1](#). See [Figure 44.1](#) for a diagram that includes the OSPI.

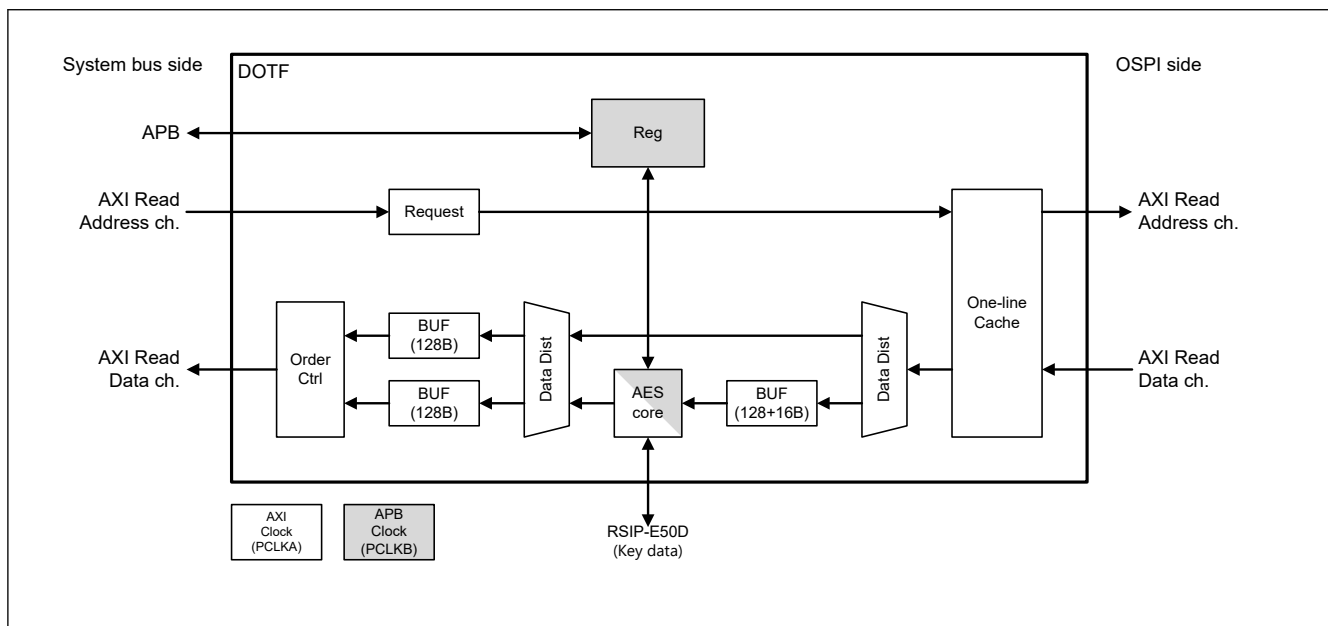


Figure 45.1 Block diagram

45.3 Register Descriptions

DOTF allows the entire area to be an accessible area, but the area where decryption processing is performed can be specified as the entire area or one contiguous area. The region can be specified by setting the start and end addresses to the CONVAREAST and CONAREAED registers, respectively.

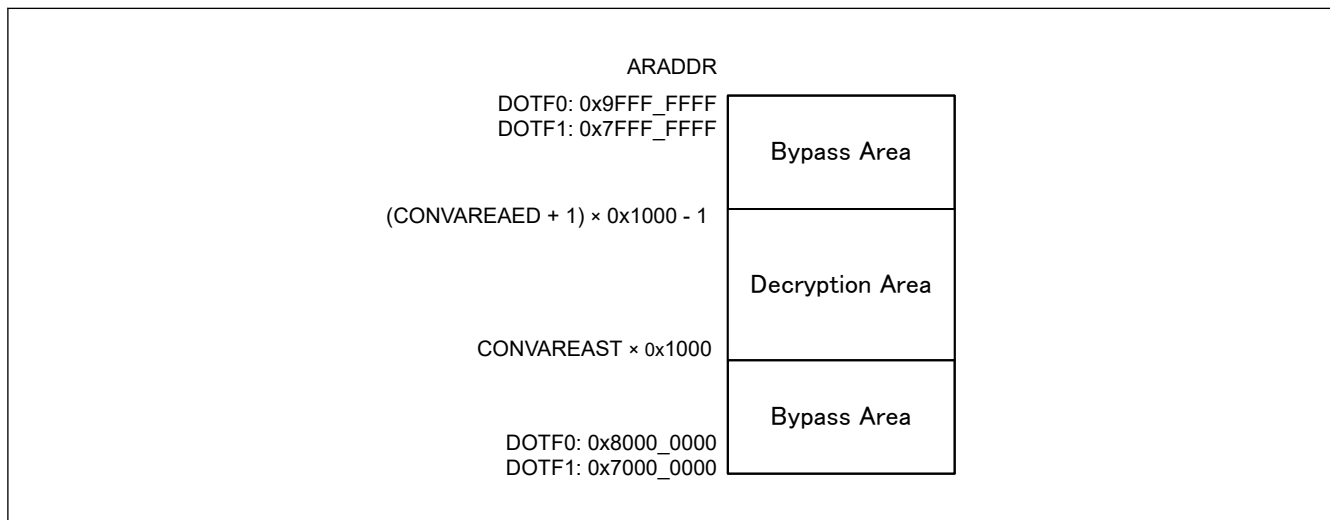
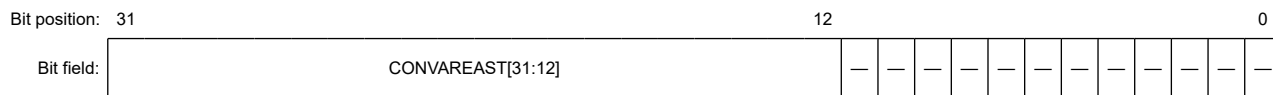


Figure 45.2 Image of decryption area setting

45.3.1 CONVAREAST : DOTF Conversion Area Start Address Register

Base address: DOTF_n = 0x4026_8800 + 0x0100 × n (n = 0, 1)
 DOTF_n_NS = 0x5026_8800 + 0x0100 × n (n = 0, 1)

Offset address: 0x00



Bit	Symbol	Function	R/W
11:0	—	These bits are read as 0.	R
31:12	CONVAREAST [31:12]	The first address of the decryption processing area. The actual address is CONVAREAST [31:12] × 0x1000.	R/W

Note: S-TYPE-3, P-TYPE-3

Specify the first address of the decryption processing area. Set before the AXI transfer request and do not make any further changes. Setting CONVAREAST[31:12] > CONVAREAED[31:12] is prohibited.

45.3.2 CONVAREAD : DOTF Conversion Area End Address Register

Base address: DOTF_n = 0x4026_8800 + 0x0100 × n (n = 0, 1)
 DOTF_n_NS = 0x5026_8800 + 0x0100 × n (n = 0, 1)

Offset address: 0x04



Bit	Symbol	Function	R/W
11:0	—	These bits are read as 1.	R

Bit	Symbol	Function	R/W
31:12	CONVAREAED [31:12]	The end address of the decryption processing area. The actual address is CONVAREAED [31:12] × 0x1000.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: Setting CONVAREAST[31:12] > CONVAREAED[31:12] is prohibited.

Specify the tail address of the decryption processing area. Set before the AXI transfer request and do not make any further changes. Setting CONVAREAST[31:12] > CONVAREAED[31:12] is prohibited.

45.4 Operation

DOTF has one AXI Read Slave Interface and one AXI Read Master Interface.

An AXI transfer request received on the AXI Slave Address channel is issued to the AXI Master's Read Address channel, and the Slave data is received from the Master's AXI Read data channel and sent to the AXI Slave Read data channel.

At this time, the necessity of decryption is determined by the requested address area, and the decrypted data is transmitted through AES core if necessary. When decryption is not required, the received response is sent without conversion.

45.5 System Flow

The initialization flow after HW reset is shown in [Figure 45.3](#).

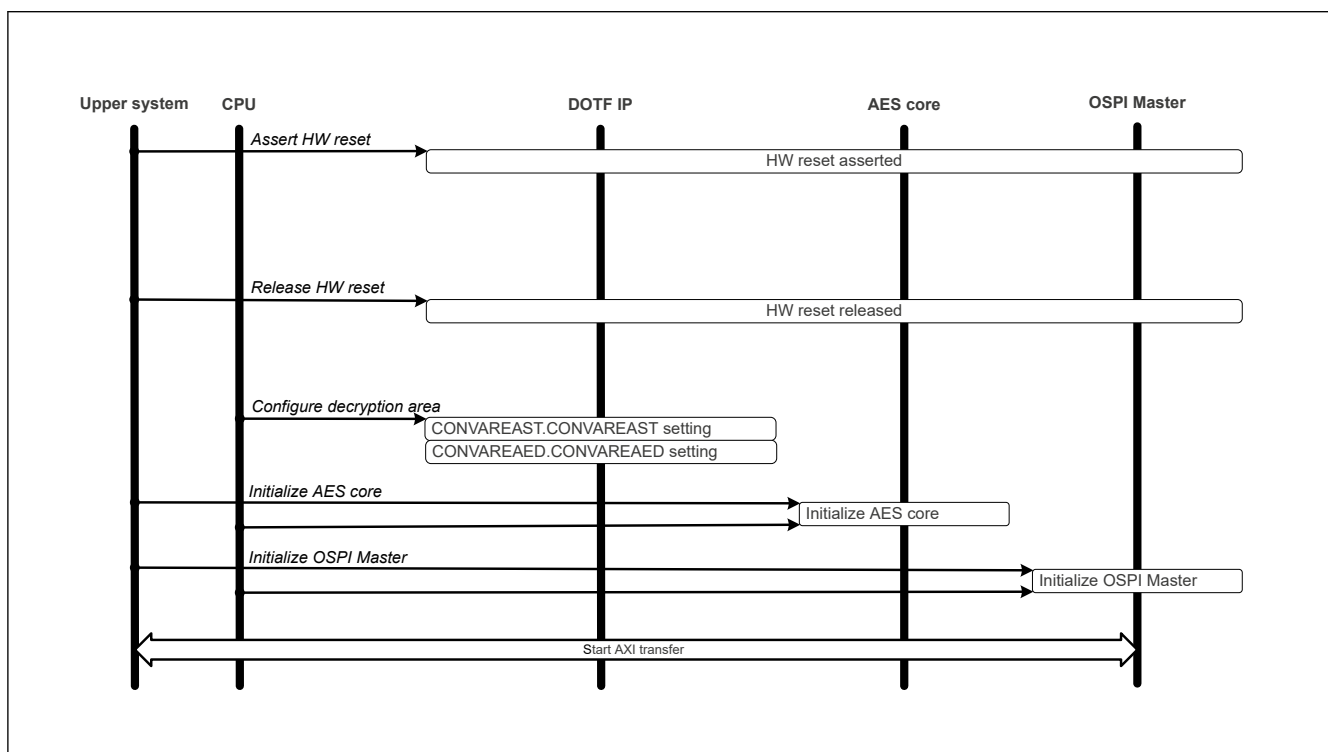


Figure 45.3 Initialization flow

45.6 Usage Notes

45.6.1 Module-stop Function

DOTF operation can be disabled or enabled using Module Stop Control Register B (MSTPCRB). The DOTF module is initially stopped after reset. Releasing the module-stop state enables access to the registers. For details, see [section 11, Low Power Mode](#).